Channel purity from a single prepared input

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On a continuous interval $0 \le f \le f\star$, the three reference readings of the first qubit admit every normalized Choi purity from J(f) to one. At f=0, measuring the complete two-qubit reference output instead bounds that same purity above by 0.812504.

What the first-qubit data imply

Here f is the probability of a specified additional outcome, $r \approx 0.4751424501$ is the available population, and $J(f)=d^2+f^2+(r-f)^2/13$ with $d=1-r$. The detector and full phase curve are fixed. All three first-qubit probabilities are set to their reference values. A pure channel has a rank-one normalized Choi matrix.

Extra-outcome probability	Compatible Choi purities
$0 \le f \le f\star \approx 0.105322985$	Exactly [J(f),1]. Both endpoints and every intermediate purity are attained.
$f\star < f \le 19r/32 \approx 0.282115830$	J(f) is attained. Distinct purities exist throughout. The maximum over all channels is unresolved here.
$19r/32 < f < r$	Purity one is excluded. A nonzero interval above J(f) is attainable. The sharp upper endpoint is unresolved here.
$f = r$	The three reference readings identify a single state with purity $d^2+r^2 \approx 0.501235795573$.

The exact $f\star$ is the positive root of $7f^2+(39d-7)f-6d(1-d)=0$. Its stated precision is a numerical evaluation. It bounds a constructive family. It supplies no impossibility result for arbitrary channels above $f\star$.

What changes when both qubits are observed

The two-qubit reference output at f=0 is mixed even though its first-qubit marginal can be reproduced by a unitary. A unitary sends a pure input to a pure joint output. Page 4 quantifies that distinction for any specified joint output. It gives a sharp interval when that output is the only datum and a valid upper bound with additional observations imposed.

A continuous set of unitary witnesses

Use output-first normalized Choi kets $|M\rangle=\mathrm{vec}(M)/2$. The match is $p=|I_4\rangle$, the other visible direction is $u=|iZ\otimes I\rangle$, and the extra outcome projects onto $v=|iX\otimes I\rangle$. A fixed effect projects onto $\sqrt{h}\,p+\sqrt{(1-h)}u$. The supplied values are $s=131/320$, $h=99/100$ and $q=81711338569043/224995905216000$. Put $k=(sh-\sqrt{(shq)})^2/[sh(1-h)]$, $d=s+k$ and $r=1-d$. The full specified phase curve fixes the p,u compression to the outer product of $w=\sqrt{s}\,p-\sqrt{k}\,u$.

Take $\cos\alpha=\sqrt{(s/d)}$, $\sin\alpha=\sqrt{(k/d)}$, and $T=\exp(-i\alpha Z/2)\otimes I$. Writing a laboratory matrix as TUT rotates the calculation into coordinates where $w/\sqrt{d}$ is $|I_4\rangle$, its null direction is $|iZ\otimes I\rangle$ and v is $|iX\otimes I\rangle$. The reference first-qubit output is diagonal in both frames. The required amplitudes in the construction below are $\sqrt{d}$ on $|I_4\rangle$, zero on $|iZ\otimes I\rangle$ and $\sqrt{f}$ on $|iX\otimes I\rangle$.

Construction

Write $\ell=r-f$, $c=\sqrt{(d+f)}$, and choose β by $c\cos\beta=\sqrt{d}$ and $c\sin\beta=\sqrt{f}$. With $P_\pm$ the first-qubit X eigenprojectors, take $U=P_+\otimes V_+ + P_-\otimes V_-$, where $V_\pm=\exp(\pm i\beta)[cI+i\sqrt{\ell}\,(n_\pm\cdot\sigma)]$ for real unit vectors $n_\pm$. Both $V_\pm$ are unitary. Taking their traces gives the three required amplitudes.

On input $|00\rangle$ the first-qubit density in the X basis has diagonals ½ and upper off-diagonal $\langle 0|V_-^\dagger V_+|0\rangle/2$. To obtain the reference output it is enough to make this overlap the real number $z=2(d+6\ell/13)-1$. After removing $\exp(2i\beta)$, the relative unitary has form $aI+i\,m\cdot\sigma$ with $a=c^2+\ell\,n_-\cdot n_+$ and $|m|^2=1-a^2$. Hence a ranges over $[1-2\ell,1]$. Jointly rotating the axes gives every imaginary part in $[-\sqrt{(1-a^2)},\sqrt{(1-a^2)}]$ for its expectation on $|0\rangle$.

> Set $a=z\cos(2\beta)$. The imaginary part $-z\sin(2\beta)$ is feasible since $|z|\le 1$. Thus this construction works precisely when $z(d-f)/(d+f)\ge 1-2\ell$. Expanding gives $7f^2+(39d-7)f-6dr\le 0$.

The polynomial is strictly increasing for $f\ge 0$ and changes sign inside $(0,r)$. Interval arithmetic puts its root strictly between 0.10532298498594176 and 0.10532298498594177. For each f in the interval, the unitary has purity one. Its overlap with the minimum state $\rho\bigstar$ is $J(f)$, since they share the sector populations. At mixture weight t on the unitary, purity is $J(f)+t^2[1-J(f)]$. This proves the entire interval.

What remains when the support changes

Let D be the thirteen-dimensional projection orthogonal to p,u,v and F=|v⟩⟨v|. The minimum state is ρ★=|w⟩⟨w|+fF+ℓD/13. Positivity annihilates its prescribed null direction in every compatible channel. The first-qubit reference rates on |00⟩ are pX=pY=½ and pZ=d+6ℓ/13. Fixing the sector populations gives Tr(ρ²)=J(f)+‖ρ−ρ★‖₂².

Support	Gram diagonal for Z and two rotated transverse axes
Interior, 0<f<r	1230/91, 97/7, 97/7
Lower endpoint, f=0	159/13, 25/2, 254/21
Upper endpoint, f=r	0, 0, ½

For each nonzero Gram entry, the reciprocal multiplies the squared deviation along that axis in the lower bound. At f=0 the two transverse axes are rotated by α relative to the laboratory axes. At f=r the only free component is a real coherence γ. The second rotated transverse probability deviation equals γ and the attainable purity is d²+r²+2γ², |γ|≤√(dr). At the reference triple γ=0.

Ambiguity and a necessary test

For any ℓ>0, six unresolved Pauli directions preserve the first bit of |00⟩ and seven flip it. Redistributing the two fixed group weights between Pauli directions leaves all first-qubit observations intact. Mixing the uniform state with a state that concentrates each weight on one direction attains the entire interval from J(f) to J(f)+72ℓ²/169. This is an attainable interval within the full model.

For a pure channel U, the visible operators act trivially on the second qubit. The residual operator has squared Frobenius norm 4ℓ. The two off-diagonal blocks of U in the second-qubit basis have equal squared norms by unitarity, so the block that flips |0⟩ has squared norm at most 2ℓ. On the pure joint output, the first and second marginals share the same eigenvalues. The first marginal must therefore have largest eigenvalue at least 1−2ℓ. At the reference readings it equals d+6ℓ/13. Purity one requires ℓ≥13r/32, equivalently f≤19r/32.

At f=0 the sharp purity floor is 0.292841628282…. The interior example at f≈0.075590554093 solves df=42ℓ²/169 and lies inside the interval on page 2. For the separate, nonreference interior triple with deviations (0.005,0.007,−0.004) in Z,X,Y order at f=r/2, the three-axis lower bound is attained at 0.336263620116203.

Complete output on one pure input

A sharp interval from the joint output state

Let Φ be any trace-preserving channel from four dimensions to four dimensions. The only datum in this theorem is its entire output σ on one specified pure input. Write P=Tr(σ²), λ=λmax(σ), and C for the trace-one Choi matrix of Φ.

> The exact interval is [(P+3/4)/16, (9+6λ+P)/16]. Both endpoints and every intermediate purity occur. Under extra detector or phase constraints these bounds remain valid, with separate attainability questions.

Lower endpoint

Choose an input basis with the measured input first. The blocks of 4C are Φ(|i⟩⟨j|). Squaring the Hilbert–Schmidt norm adds their squared norms. The first diagonal block contributes P, each of the other three density matrices contributes at least ¼, and all other contributions are nonnegative. Measure the input in this basis, send the first outcome to σ and the other outcomes to $I_4/4$. This channel attains the lower endpoint.

Upper endpoint

Take a Stinespring isometry. For each basis input j let τj be the environmental marginal of its pure joint output. Then τ_0 has the nonzero spectrum of σ. The maximally entangled input purifies C with environmental marginal $(\tau_0+\tau_1+\tau_2+\tau_3)/4$. Writing $\omega=(\tau_1+\tau_2+\tau_3)/3$ gives 16 $\mathrm{Tr}(C^2)=P+6\mathrm{Tr}(\tau_0\omega)+9\mathrm{Tr}(\omega^2)\le P+6\lambda+9$.

In a Schmidt purification of σ choose a system and environment Schmidt pair for λ. For the other three inputs choose orthonormal system vectors perpendicular to this system vector, all paired with that same environment vector. The four images are orthonormal, so they define an isometry. Here ω is pure in the top eigenspace of τ_0 and the upper bound is attained. Interpolation between the two channels keeps σ fixed and, by continuity, attains all intermediate purities.

Application to the fixed phase experiment

For ρ★ the full output on |00⟩ is diagonal with weights $d+2\ell/13$, $4\ell/13$, $f+3\ell/13$, $4\ell/13$ in binary order. At f=0 its purity ceiling from the joint output alone is approximately 0.812503777157. The first-qubit data alone at that f permit purity one. At f=r/2 the complete-output ceiling is approximately 0.798737595195. All these joint reference outputs are mixed, including both endpoints. No unitary reproduces any of them on the pure input.

Purity range at f=0 with the phase observations fixed



Solid green denotes attained purities. Dots mark values below the joint-output ceiling whose attainability is unresolved in the fixed phase model.

How much of the channel is determined?

On the allowed support, every phase effect is a multiple of the same rank-one projector. Extra phase settings consequently cannot recover independent directions in the channel. The table gives exact real dimensions of the local set of channels at the reference state. “Joint” fixes the complete two-qubit output on the same input. Unitality means $\Phi(I)=I$.

Extra outcome	First qubit	First, unital	Joint	Joint, unital
$0<f<r$	204	189	192	177
$f=0$	176	161	164	149

To certify the ranks, use the rotated Pauli frame of page 2 and compress the constraint matrices to the support. Include sixteen reference-marginal rows, the independent sector-population rows and three or fifteen output rows. Add fifteen output-marginal rows for unitality. Clearing the dyadic denominators gives integer rows. Elimination modulo 101 gives full row rank in each case. The reference state is positive definite on its support, so every sufficiently small tangent perturbation is physically allowed. These dimensions apply to the specified reference outcomes.

Even among mixtures of unitaries, the four output groups of the unresolved Pauli directions have sizes 2,4,3,4 on $|00\rangle$. Holding their group weights fixed gives a product of simplexes of dimension nine. At $f=r/2$ it realizes purities from approximately 0.336257079801 to 0.346943960175 with identical joint output. Thus the upper bound on page 4 can exclude purity one without determining the channel.

What an observation could establish

A mixed joint output for an independently pure input excludes a unitary channel on that input path. For measured data, maximize the upper bound over a simultaneous confidence region for the joint output. Additional phase and detector information can restrict the feasible set further. An instrument conditioned on history 111 can be trace-decreasing. Its success probability and transfer across preparations need independent characterization before applying a normalized-channel theorem. No matched hardware trials are part of these calculations.

Mathematical inputs · Endpoint constraints in three-axis channel tomography.

Mathematical inputs · Operational identification and observation fibers.

Watrous, The Theory of Quantum Information, ch. 2 · channel representations.

Ziman (2008) · incomplete process tomography.

Keith et al. (2018) · compatible state sets and bounds.

White et al. (2022) · experimental multitime process characterization.